

Colab → GitHub: Step-by-Step Guide (Word)

This document contains detailed, copy-paste ready, step-by-step instructions to: set up and run your Google Colab notebook, save and sync it to GitHub, add an "Open in Colab" badge, create a `requirements.txt`, and add a GitHub Actions workflow that executes your notebook automatically. The instructions are suitable for saving as a Microsoft Word (.docx) file.

Quick Overview

1. Create or update `requirements.txt` in your repository.
 2. Open the notebook in Google Colab and install dependencies.
 3. Download the dataset using `kagglehub` (or commit a sample dataset to the repo).
 4. Inspect, clean, analyze, and visualize the data in Colab.
 5. Save notebook changes back to GitHub.
 6. Add an "Open in Colab" badge to the README.
 7. Add a GitHub Actions workflow to execute the notebook on push.
 8. Troubleshoot common issues.
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1. Create `requirements.txt`

Create a file named `requirements.txt` in the root of your GitHub repository. Add at minimum the following lines (you can pin versions if desired):

```
pandas
numpy
matplotlib
plotly
seaborn
kagglehub
nbformat
nbconvert
papermill
tensorflow
```

Save and commit this file to your repo.

2. Open the Notebook in Google Colab and Install Dependencies

1. Open your notebook in Colab by using the URL pattern:

```
https://colab.research.google.com/github/<username>/<repo>/blob/main/<path-to-notebook>.ipynb
```

1. In Colab, add a new top cell and run the following to install dependencies:

```
# Install (only required per session)
!pip install --upgrade pip
!pip install -r /content/requirements.txt || pip install pandas numpy matplotlib
plotly seaborn kagglehub papermill nbconvert tensorflow
```

Notes: - If you have `requirements.txt` in the repo and you opened the notebook via GitHub, you can fetch it. If Colab can't access the file, the fallback `pip install` will ensure required packages are installed.

3. Download Dataset from Kaggle (in Colab)

Use `kagglehub` to download the dataset. Add and run this cell:

```
import kagglehub

# download the ice-cream dataset
path = kagglehub.dataset_download("tysonpo/ice-cream-dataset")
print("Path to dataset files:", path)
```

If `kagglehub` asks for authentication outside of Colab, follow the provided link and authenticate. In Colab, authentication is often seamless.

Alternative: If Kaggle access is problematic for the CI workflow, pre-download and commit a small, sanitized CSV to the repo at `data/ice_cream_sample.csv` so GitHub Actions can run without needing Kaggle auth.

4. Inspect and Load Data

Add a cell to inspect dataset files and load the main CSV:

```
import os, glob, pandas as pd

dataset_dir = path # returned from kagglehub
print("Files in dataset dir:")
for f in glob.glob(os.path.join(dataset_dir, "*")):
```

```

    print("-", f)

csv_candidates = glob.glob(os.path.join(dataset_dir, "*.csv"))
print("CSV candidates:", csv_candidates)

# load the first CSV found
if csv_candidates:
    df = pd.read_csv(csv_candidates[0])
else:
    raise FileNotFoundError("No CSV files found in dataset directory")

# quick view
print(df.shape)
df.head()

```

5. Clean & Explore

Add cells for basic cleaning and exploratory analysis:

```

# basic cleaning and dtype normalization
df = df.copy()
print(df.isna().sum())

# convert rating if exists
if 'rating' in df.columns:
    df['rating'] = pd.to_numeric(df['rating'], errors='coerce')

# show summary stats
print(df.describe(include='number').T)

```

6. Analysis: Top Flavors by Average Rating

```

if 'flavor' in df.columns and 'rating' in df.columns:
    top_flavors = df.groupby('flavor')
    ['rating'].mean().sort_values(ascending=False).head(10).reset_index()
    top_flavors.columns = ['flavor', 'avg_rating']
    display(top_flavors)
else:
    print("Columns 'flavor' and 'rating' not found. Columns:", list(df.columns))

```

7. Visualization (Interactive Plotly)

```
import plotly.express as px

fig = px.bar(top_flavors, x='flavor', y='avg_rating',
             title='Top 10 Ice Cream Flavors by Average Rating',
             labels={'avg_rating': 'Average Rating', 'flavor': 'Flavor'})
fig.update_layout(xaxis_tickangle=-45)
fig.show()
```

Plotly charts render interactively in Colab.

8. Save Results and Push to GitHub

To save analysis output locally in Colab:

```
out_path = "/content/top_flavors_summary.csv"
top_flavors.to_csv(out_path, index=False)
print("Saved:", out_path)
```

Push notebook changes to GitHub - Use **Colab UI**: `File → Save a copy in GitHub` and provide a commit message. - Or use `git` on the command line if you prefer to set credentials in Colab (advanced).

9. Add "Open in Colab" Badge to README

Edit `README.md` and add:

```
[![Open In Colab](https://colab.research.google.com/assets/colab-badge.svg)]
(https://colab.research.google.com/github/Sneha2025L/SnehaGoogleColabFiles/blob/
main/notebooks/IcrecreamAnalysis/
Copy_of_Getting_started_with_google_colab_ai.ipynb)
```

Adjust the URL to your notebook path.

10. GitHub Actions: Auto-Execute Notebook on Push

Create `.github/workflows/execute-notebook.yml` with the content below. This executes the notebook using `papermill` and uploads the executed notebook as an artifact.

```

name: Execute Notebook

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  run-notebook:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repo
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v4
        with:
          python-version: '3.10'

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt
          pip install papermill nbformat nbconvert

      - name: Execute notebook with papermill
        run: |
          mkdir -p ./executed
          papermill notebooks/IcrecreamAnalysis/
Copy_of_Getting_started_with_google_colab_ai.ipynb executed/
Copy_of_Getting_started_with_google_colab_ai-executed.ipynb --log-output

      - name: Upload executed notebook as artifact
        uses: actions/upload-artifact@v4
        with:
          name: executed-notebook
          path: executed/Copy_of_Getting_started_with_google_colab_ai-
executed.ipynb

```

Important: If your notebook depends on Kaggle downloads via interactive auth, GitHub Actions may fail. Option A: commit a sample dataset to the repo (`data/` folder). Option B: use GitHub secrets and Kaggle API key (more advanced).

11. Troubleshooting

- **Kaggle auth errors in CI:** pre-download dataset and commit a sanitized CSV to `data/`.
 - **Missing packages in Actions:** ensure `requirements.txt` includes all dependencies.
 - **Notebook path mismatch:** update the path in the workflow to match your repo layout.
 - **Large outputs causing workflow timeouts:** clear large plot outputs before saving or use smaller sample data for CI.
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12. How to Save This as a Word (.docx) File

1. Open this document in Colab or any Markdown editor that supports export (or copy the content into Microsoft Word).
2. In Microsoft Word: paste the content and save as `Colab_GitHub_Steps.docx`.
3. Alternatively, use a Markdown-to-docx converter (e.g., Pandoc):
4. Install Pandoc and run:

```
pandoc -o Colab_GitHub_Steps.docx Colab_GitHub_Steps.md
```

13. Final Checklist (Before pushing to GitHub)

- ☐ `requirements.txt` added and committed
 - ☐ Notebook path in workflow is correct
 - ☐ README badge added
 - ☐ Small sample dataset added to `data/` for CI (optional but recommended)
 - ☐ Push and verify GitHub Actions run successfully
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If you want, I can now: - Export this document into a downloadable Word `.docx` file for you, or - Create the exact `execute-notebook.yml` and `requirements.txt` files tailored to your repo path and commit messages.

Which would you like next?